

Living Typography: Robotically Printing a Living Typeface

Brendan Harmon*
College of Art and Design
Louisiana State University
Baton Rouge, Louisiana, USA
baharmon@lsu.edu

Hunter Gilbert
College of Engineering
Louisiana State University
Baton Rouge, Louisiana, USA
hbgilbert@lsu.edu

Hye Yeon Nam*
College of Art and Design
Louisiana State University
Baton Rouge, Louisiana, USA
hyenam@lsu.edu

Nasrin Iravani
College of Art and Design
Louisiana State University
Baton Rouge, Louisiana, USA
nirava4@lsu.edu



Figure 1: A robotically planted living typeface.

ABSTRACT

Artists and designers have been exploring how robotics can be used to interact with our environment in new ways. Robots connect computational design processes with the physical environment, making digital interaction with nature possible. We present a robotic process for planting that enables the computational design of landscapes. We demonstrate how robotic planting can be used for generative art and design by creating a living typeface grown from seed. The robot draws a message by 3-dimensionally (3D) printing a blend of planting media and seeds. When the seeds germinate, the glyphs emerge from their substrate in a flush of green. The letterforms become dynamic living organisms. Artistic agency shifts from the artist to nature.

*Both authors contributed equally to this research.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

© 2022 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3519894>

CCS CONCEPTS

• Human-centered computing → Visualization; • Hardware → Emerging interfaces.

KEYWORDS

robotics, digital craft, nature, typography

ACM Reference Format:

Brendan Harmon, Hye Yeon Nam, Hunter Gilbert, and Nasrin Iravani. 2022. Living Typography: Robotically Printing a Living Typeface. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3491101.3519894>

1 INTRODUCTION

As interfaces between the digital and the physical, robotics have the potential to dramatically transform how we interact with our environment. While much work in robotics has focused on automating tasks, creative applications of robotics can enable novel interactions. Artists and designers have been experimenting with environmental applications for robotics such as interactive performances, generative fabrication, and non-standard construction.



Figure 2: A series of seven silhouettes showing the changing outline of the letter C as it grows over the course of two weeks.

Artists, for example, have used robots to enable online communities to plant and tend gardens [10, 12], to autonomously care for gardens using machine learning [11], and to plant fields with algorithmically generated patterns of crops, herbs, and flowers [13]. Architects have used robotic fabrication and assembly processes – such as weaving tensile structures [5, 18], automated brick laying [2, 6, 9], stone cutting [3], assembling timber frames [7, 19], constructing complex formwork [4, 8], and 3D printing metal [16] and concrete structures [20] – to construct the built environment in new ways. Landscape architects have used autonomous excavators to computationally construct landscapes [15, 17]. We present a novel process for robotic planting (Figure 3) and demonstrate how it can be used for generative art with a living typeface (Figure 1). Our concept of living typography differs from generative typography [1, 14] in that creative agency is shared with nature, rather than just the artist and algorithm. This planting process could be scaled up with field robotics for applications such as generative design in landscape architecture and automation in agriculture.

2 PROCESS

In Living Typography letterforms are drawn with robotically seeded plants. A message is written in seed by a robot and grows into a unique living form of typography, until it is harvested and eaten. The design is 3D printed by a collaborative robotic arm mounted with an extruder (Figure 3). The extruder prints a blend of planting media and seeds, creating 3D forms – here, letterforms. As the seeds germinate, seedlings sprout out of the planting media, their roots growing into the ground below. The design transforms from a clay body into a profusion of green plants. The topographic typeface becomes a community of living organisms, competing for light, water, and nutrients. Its shape and color change as the plants grow (Figure 4). As the shoots grow and spread, the letterforms become bolder and more vibrant, with leaves adding weight. The minute detail of the leaves jostling for space and light creates a complex form with an irregular, convoluted edge (Figure 2). Eventually the letterforms become obscured as the plants spread into the negative space between the strokes, filling the counters. When the letters disappear in a field of green, the plants are ready for harvesting and serving as microgreens.



Figure 3: Robotic planting process

3 CONCLUSION

Living Typography demonstrates how generative art can interface with nature through robotics, distributing agency between artist, algorithm, machine, and nature. In this demonstration the 3D printed letters become a living form of typography in which nature has creative agency, transforming a standard typeface into a unique, evolving form. Integrating sensors into the robotic system could enable real-time interactions with environment. The seed printing process demonstrated here could be scaled up with field robotics to plant entire landscapes, enabling the algorithmic design of ecosystems for the sake of art and ecology.



Figure 4: Evolution of 3D printed letters over the course of two weeks.

ACKNOWLEDGMENTS

This work was funded by a faculty research grant from Louisiana State University's Office of Research and Economic Development and supported by the Coastal Sustainability Studio.

REFERENCES

- [1] Y. Ahn and V. Cordova. 2008. *Type + Code: Processing for Designers*. Center for Design Thinking, Maryland Institute College of Art.
- [2] Frederico Augugliaro, Sergei Lupashin, Michael Hamer, Cason Male, Markus Hehn, Mark W Mueller, Jan Sebastian Willmann, Fabio Gramazio, Matthias Kohler, and Raffaello D'Andrea. 2014. The Flight Assembled Architecture installation: Cooperative construction with flying machines. *IEEE Control Systems Magazine* 34, 4 (2014), 46–64. <https://doi.org/10.1109/MCS.2014.2320359>
- [3] Mark Burry. 2016. *Robots at the Sagrada Familia Basilica: A Brief History of Robotised Stone-Cutting*. Springer International Publishing, Cham, 2–15. https://doi.org/10.1007/978-3-319-26378-6_1
- [4] Ron Culver, Julia Koerner, and Joseph Sarafian. 2016. *Fabric Forms: The Robotic Positioning of Fabric Formwork*. Springer International Publishing, Cham, 106–121. https://doi.org/10.1007/978-3-319-26378-6_8
- [5] Niccolò Dambrosio, Christoph Zechmeister, Serban Bodea, Valentin Koslowski, Marta Gil-Pérez, Bas Rongen, Jan Knippers, and Achim Menges. 2019. Buga Fibre Pavilion: Towards an architectural application of novel fiber composite building systems. *Ubiquity and Autonomy - Paper Proceedings of the 39th Annual Conference of the Association for Computer Aided Design in Architecture, ACADIA 2019* (2019), 140–149. http://papers.cumincad.org/cgi-bin/works/paper/acadia19_140
- [6] Kathrin Dörfler, Timothy Sandy, Markus Gifftthaler, Fabio Gramazio, Matthias Kohler, and Jonas Buchli. 2016. *Mobile Robotic Brickwork*. Springer International Publishing, Cham, 204–217. https://doi.org/10.1007/978-3-319-26378-6_15
- [7] Philipp Eversmann, Fabio Gramazio, and Matthias Kohler. 2017. Robotic prefabrication of timber structures: towards automated large-scale spatial assembly. *Construction Robotics* 1, 1-4 (2017), 49–60. <https://doi.org/10.1007/s41693-017-0006-2>
- [8] Nadja Gaudillière, Romain Duballet, Charles Bouyssou, Alban Mallet, Philippe Roux, Mahriz Zakeri, and Justin Dirrenberger. 2019. Large-Scale Additive Manufacturing of Ultra-High-Performance Concrete of Integrated Formwork for Truss-Shaped Pillars. In *Robotic Fabrication in Architecture, Art and Design 2018*, Jan Willmann, Philippe Block, Marco Hutter, Kendra Byrne, and Tim Schork (Eds.). Springer International Publishing, Cham, 459–472.
- [9] Markus Gifftthaler, Timothy Sandy, Kathrin Dörfler, Ian Brooks, Mark Buckingham, Gonzalo Rey, Matthias Kohler, Fabio Gramazio, and Jonas Buchli. 2017. Mobile robotic fabrication at 1: 1 scale: the in situ fabricator. *Construction Robotics* 1, 1-4 (2017), 3–14. <https://doi.org/10.1007/s41693-017-0003-5>
- [10] Ken Goldberg. 1995. *The Telegarden*. Retrieved January 12, 2022 from <https://goldberg.berkeley.edu/garden/Ars/>
- [11] Ken Goldberg. 2020. *AlphaGarden*. Retrieved January 12, 2022 from <http://alphagarden.org>
- [12] K. Goldberg, S. Gentner, C. Sutter, and J. Wiegley. 2000. The Mercury Project: a feasibility study for Internet robots. *IEEE Robotics Automation Magazine* 7, 1 (2000), 35–40. <https://doi.org/10.1109/100.833573>
- [13] Benedikt Groß. 2013. *Avena+ Test Bed: Agricultural Printing and Altered Landscapes*. Retrieved January 12, 2022 from <https://benedikt-gross.de/projects/avena-test-bed-agricultural-printing-and-altered-landscapes>
- [14] B Groß, H Bohnacker, J Laub, C Lazzeroni, J Lee, N Poldervaart, and M Frohling. 2018. *Generative Design: Visualize, Program, and Create*. Princeton Architectural Press.
- [15] Ilmar Hurkxkens. 2020. *Robotic Landscapes: Topological Approaches to Terrain, Design, and Fabrication*. Ph. D. Dissertation. ETH Zurich, Zurich. <https://doi.org/10.3929/ethz-b-000451100>
- [16] Joris Laarman Lab. 2018. *MX3D Bridge*. Retrieved January 12, 2022 from <https://www.jorislarmann.com/work/mx3d-bridge/>
- [17] D. Jud, P. Leemann, S. Kerscher, and M. Hutter. 2019. Autonomous Free-Form Trenching Using a Walking Excavator. *IEEE Robotics and Automation Letters* 4, 4 (2019), 3208–3215.
- [18] Marshall Prado, Moritz Dörstelmann, Achim Menges, James Solly, and Jan Knippers. 2017. *Elytra filament pavilion: Robotic filament winding for structural composite building systems*. UCL Press, 224–231. <https://doi.org/10.2307/j.ctt1n7qkg7.35>
- [19] Andreas Thoma, Arash Adel, Matthias Helmreich, Thomas Wehrle, Fabio Gramazio, and Matthias Kohler. 2019. Robotic Fabrication of Bespoke Timber Frame Modules. In *Robotic Fabrication in Architecture, Art and Design 2018*, Jan Willmann, Philippe Block, Marco Hutter, Kendra Byrne, and Tim Schork (Eds.). Springer International Publishing, Cham, 447–458.
- [20] US Army Corps of Engineers. 2017. *Automated Construction of Expeditionary Structures (ACES)*. Retrieved January 12, 2022 from <https://www.erdc.usace.army.mil/Media/Fact-Sheets/Fact-Sheet-Article-View/Article/1290247/automated-construction-of-expeditionary-structures-aces/>